

Multi-scale intrinsic dimension of text: estimator calibration and a controlled intervention study

A research note extending the qPHD method to ModernBERT embeddings

Abstract. The persistent-homology dimension of a text's token embedding cloud can be measured at a chosen scale by discarding a fraction q of the minimum spanning tree's edges before summing them. We port this estimator to ModernBERT embeddings, calibrate it, and use it to ask which properties of a text causally determine the dimension it measures. Calibration proves necessary rather than cosmetic: the published edge-counting rule introduces a quantisation artefact that is invisible on a coarse q grid, and the estimator's value depends strongly on the number of tokens sampled, a dependence no configuration removes. A reconfigured estimator reduces the relative noise of a single measurement from 0.114 to 0.026. Applied to 3000 texts across four genres and five sources, it places every generator above human text at every q , with separability maximised near $q = 0.3$ rather than at $q = 0$, i.e. above the untrimmed estimator. A controlled intervention study over 31 perturbations isolates four causal factors: lexical diversity, word order within a sentence, topic variety and idea count. Two methodological findings constrain how such studies must be run: every LLM rewrite carries a shared component that raises dimension regardless of the instruction, and an instruction that is only weakly executed yields not a null result but a result of the wrong sign.

1. Method

A text is passed through ModernBERT-base, giving one 768-dimensional point per token. The intrinsic dimension of that cloud is estimated from the growth of the total minimum-spanning-tree edge length with the number of points, $S(n) \propto n^b$ with $b = 1 - \alpha/d$, so that $d = \alpha/(1 - b)$, with $\alpha = 1$ throughout. Discarding a fraction q of the edges before summing selects a scale: short edges carry local structure, long edges global structure. Three trimming modes are used — q_small discards the q shortest edges and so observes the coarse scale, q_large discards the q longest and observes the fine scale, and $q0.5_range$ trims from both sides and keeps the central half, a sliding window.

One measurement fixes L tokens, forms a grid of subsample sizes n , draws 32 subsamples at each n , builds an MST per subsample, trims, sums, averages, and fits a straight line to $\log S$ against $\log n$. The MST is built once per subsample and reused for every q and mode, since trimming is only a different sum over one sorted edge list. Two sources of randomness make a single measurement a random variable: which tokens are drawn from the text, and which subsamples are drawn within.

1.1 Two measures of estimator quality

Measurement noise is separated from genuine text-to-text differences by a one-way random-effects decomposition over N texts and k seeds, $d_{ij} = \mu + a_i + e_{ij}$. The within-text variance is estimated inside each text, where a_i is constant and cancels, and averaged across texts; the variance of the per-text means is inflated by σ_w^2/k , the standard error of the mean, and is debiased by subtracting it. Noise is reported relative to d , since d spans a factor of five across q .

The second measure is the residual of the log-log fit, which lives in log space and is therefore already relative. The two are independent, and an estimate good by one can be bad by the other. Because $d = \alpha/(1 - b)$, a shift in the slope propagates to d amplified by a factor of d itself: at $d = 21$ a slope error of 0.005 gives a 10% error in d , where at $d = 3.5$ the same slope error gives 1.7%. Consequently the larger the measured dimension, the less it can be trusted, and a good fit does not compensate.

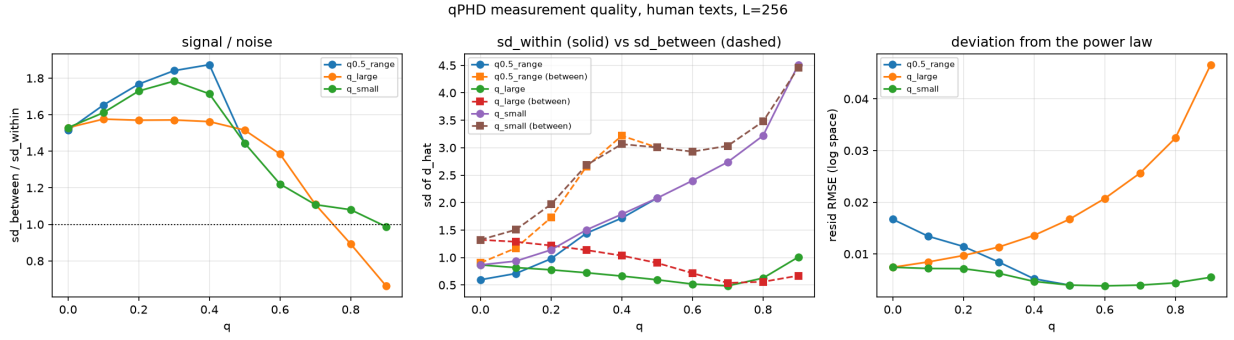


Figure 1. Measurement quality against q , human texts, $L = 256$, under the original configuration. Left: ratio of text-to-text differences to measurement noise. Centre: the two components separately. Right: deviation from the power law. The two criteria disagree by construction — at high q_{small} the power law holds best of anywhere in the table (0.3%) while the noise is at its maximum, because there d is large and slope error is amplified accordingly; at high q_{large} the model itself breaks down (4.3%).

2. Estimator calibration

2.1 A quantisation artefact in the published rule

The reference implementation discards $k = \text{floor}(q \cdot m)$ edges, where $m = n - 1$ varies across the fit grid. The realised trimmed fraction k/m therefore differs between subsample sizes: at the smallest subsample floor rounds q down by up to $1/m$, less is discarded, S is inflated there, and the fitted slope tilts. The result is a saw-tooth in $d(q)$ with troughs at $q = 0.05, 0.15, 0.25, 0.35$ and 0.45 , of amplitude 0.5 – 0.9 in d . It is invisible on the original grid of step 0.1 , which lands only where floor is exact for every m , and it reproduces on uniform 8-dimensional synthetic data whose true curve must be smooth.

Two remedies were compared against giving the boundary edge a fractional weight $1 - \text{frac}(q \cdot m)$. Rounding instead of flooring removes only a floating-point special case: on a non-aligned grid it gives a roughness of 2.235 against 2.497 for flooring, where fractional weighting gives 0.127 , because the artefact arises from $k/m \neq q$ rather than from the sign of the error. Aligning the grid so that every m is a multiple of 20 makes $q \cdot m$ an integer for the whole grid and removes the need to round at all; on such a grid rounding and fractional weighting agree exactly, which also establishes that fractional weighting is the correct continuation of exact integer trimming rather than a heuristic. Both are adopted: the grid is aligned, and fractional weighting is retained as a guard for q off the grid and for binary arithmetic, where for instance $0.35 \cdot 180$ evaluates to 62.99999999999999 and floors to 62 instead of 63 .

2.2 Configuration

The remaining hyperparameters were selected by a paired sweep on common texts. Relative noise of a single measurement falls from 0.114 under the published settings to 0.026 .

Parameter	Value	Evidence
Subsampling	without replacement	bootstrap leaves $\sim 37\%$ duplicate points at $n = L$, biasing d up 12 – 18%
Pool	$2L$ tokens	halves noise, leaves d unchanged (7.88 vs 7.89)
Replicates	32	$0.114 \rightarrow 0.076$; 64 gives 0.066 at twice the cost
n grid	aligned, $41 \dots 201$	grid shape is immaterial; grid range is not
Trimming	fractional	see 2.1
L	201 , absolute	a fractional L reimports length dependence
L against q	independent	rescaling costs the q axis its meaning

2.3 Dependence on the number of sampled tokens

The estimate follows $d \propto L^\gamma$ with $\gamma \approx -0.21$ at $q = 0$ and up to -0.43 at $q_{small} = 0.3$. The reconfigured estimator barely changes the exponent (-0.24 and -0.49 before), which identifies this as structural rather than an artefact, and no convergence is reached within the accessible range: each doubling of L removes a further 8 – 10% . The interpretation is that d is estimated over the window n in $[0.2L, L]$, and different L are different ranges of scale; since qPHD exists precisely to measure how dimension varies with scale, independence of the measurement

window should not be expected. L must therefore be matched exactly, and absolute values of d carry no meaning without it. One near-invariant point exists, $q_{\text{large}} \approx 0.6$, with $\gamma = -0.13$ and $d(768)/d(256) = 1.01$.

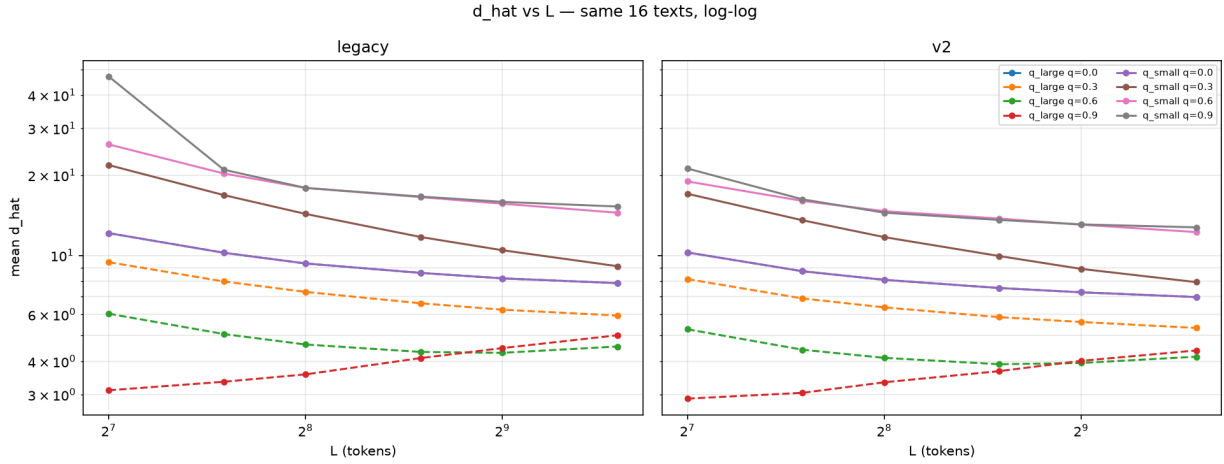


Figure 2. d against L on identical texts, log-log, under the published settings (left) and the reconfigured estimator (right). The slopes are nearly unchanged, and no plateau appears. $q_{\text{large}} = 0.9$ runs opposite in sign to everything else, so the direction of the dependence itself turns somewhere between q_{large} 0.6 and 0.9.

With L fixed as an absolute count, the original length of the text no longer influences the result: regressing $\log d$ on $\log(\text{token count})$ within genre over 160 texts of 322–881 tokens gives slopes between -0.014 and $+0.031$, indistinguishable from zero. Setting L as a fraction of the text instead reimports the dependence in full, with slopes of -0.13 to -0.56 , as it must: if $d \propto L^{-0.2}$ and L is proportional to length, then d becomes a function of length.

Scaling L with q so that the retained edge count is constant — the obvious remedy for the growing noise at high q — was tested and rejected. It achieves its aim, noise ceasing to grow and instead falling from 0.070 to 0.037, but the problem it addresses has already been removed: at fixed $L = 228$ the reconfigured estimator stays below 0.068 across the whole range, against 0.228 originally. Meanwhile the rescaling bends the $d(q)$ curve through the known L -dependence, and correcting the scaled arm back to a common L by $L^{0.21}$ reproduces the fixed- L arm (16.03 against 16.10 at $q = 0.6$), confirming that the difference carries no additional information.

3. Genres and generators

The calibrated estimator was applied to four genres and five sources, 150 texts each, 3000 texts in total. Genre ordering among human texts is stable at low q — xsum 9.07, amazon_reviews 8.40, writingprompts 8.14, academic_abstracts 7.46 — but the curves cross near $q = 0.75$: academic_abstracts starts lowest and reaches 19.15 by $q = 0.9$, above all others, while writingprompts saturates near 15 and then declines. A single value of q is therefore insufficient to characterise a genre; the shape of the curve carries information that no single point does.

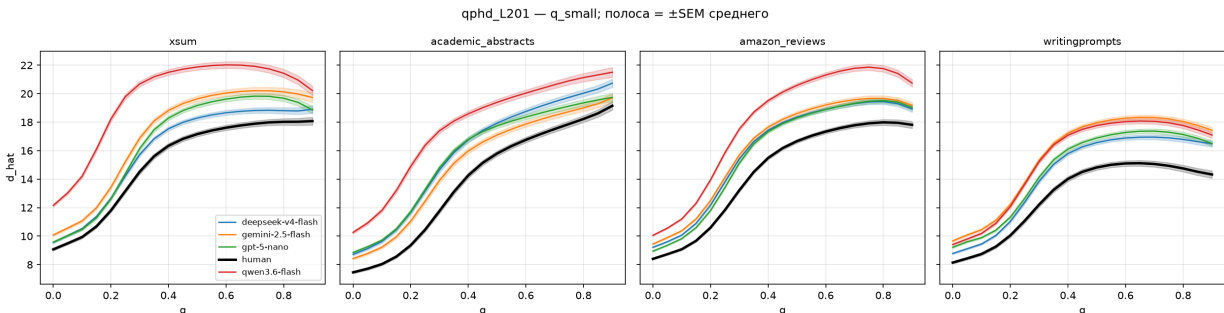


Figure 3. qPHD against q by source within each genre, q_{small} . Human text (black) lies below every generator in every genre, at every q and in all three trimming modes, without exception. Bands are standard errors of the mean, which at $n = 150$ are about twelve times narrower than the text-to-text spread; they show that the group averages are well separated, not that individual texts are.

Separability of individual texts is measured by the ROC-AUC of d between human and each model. Averaged over genres and models in the $q0.5_range$ mode it rises from 0.775 at $q = 0$ to 0.831 at $q = 0.35$ before falling to

0.792 at $q = 0.5$, so the trimmed estimator is more informative than the untrimmed one, which corresponds to $q = 0$. Effect sizes behave the same way: Cohen's d for qwen3.6-flash rises from 2.23 at $q = 0$ to 2.44 at $q = 0.3$ and halves again by $q = 0.9$. The largest value observed is 0.994 for qwen3.6-flash in xsum.

AUC here is a rank statistic of two samples rather than the accuracy of a fitted classifier, so no held-out data is required for it to be unbiased. The *choice* of the best q is made on the same data, and was therefore checked by split-half selection over 20 repeats: the selection bias averages +0.007 AUC and reaches +0.032 at most, the curve being smooth in q so that the effective number of independent attempts is far below the 57 grid points. Headline values are unchanged, and the third decimal is accordingly not reported.

qwen3.6-flash separates about twice as strongly as the others, while deepseek-v4-flash, gpt-5-nano and gemini-2.5-flash form a tight group that these curves barely distinguish. The measure captures machine authorship well and the identity of the model poorly. Note that generator texts open with a human prefix of roughly sixty words that the model was asked to continue; it was retained, as the dataset is built that way, and works against the reported separation.

4. Which properties causally determine dimension

Thirty-one perturbations were applied to 50 texts per genre and compared against an unperturbed arm on the same text, so that between-text variance — the larger of the two variance components — drops out. Ten perturbations are mechanical and involve no model; twenty-one are LLM rewrites, 4000 calls in total, each holding length within about 10% and changing one named property.

4.1 Fidelity checking is not optional

Every rewrite was verified against a metric tracking its target property before any dimension was computed, the criterion being the share of texts moved in the intended direction, for which 0.5 is chance rather than half success. Two rewrites initially failed this check, and the consequence was not a null result but a result of the wrong sign. Instructions phrased in relative terms — 'more' or 'fewer' ideas — were satisfied by rewriting in general: both arms then raised d , because every LLM rewrite raises d on its own, as Section 3 shows independently. Restating the target as an absolute property checkable sentence by sentence, and moving to a stronger model, raised execution from 0.49 to 0.80 and reversed the measured effect from +19.2% to -40.0%. The topic pair failed and was repaired identically. Two cases out of two.

Two metric choices had to be discarded along the way. A word-count measure of idea density rises when a text is shortened, by Heaps' law, and again when it is reworded, so it reported the opposite of the truth; it was replaced by the mean pairwise distance between sentence embeddings. Those raw cosines sit near 0.93 because the embedding space is anisotropic, leaving almost no dynamic range, so the vectors must be centred first — without centring every manipulation looks like a no-op.

4.2 Results

Evidence is graded by three criteria. A *reversal*, the two arms moving d in opposite directions, cannot be produced by a component common to both arms and so is evidence that the named property is responsible. *Replication* across two independent rewriting models guards against a quirk of one model's prose. A *mechanical control*, a direct manipulation with no model involved, is the strongest form available. Effect size is Cohen's d_z of the paired differences; under normality d_z maps onto the share of texts moving one way, and the mapping holds closely in these data ($d_z = 2.43$ predicts 99.3% and 99.5% is observed).

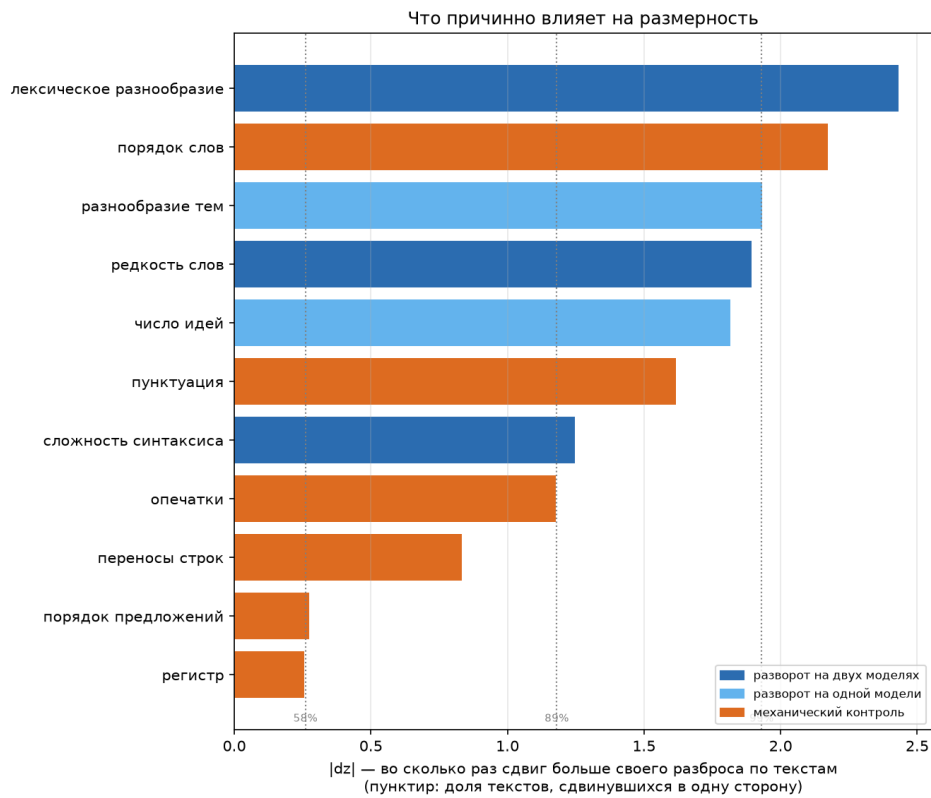


Figure 4. Properties ranked by $|dz|$, coloured by the kind of evidence available. Dotted lines mark the share of texts moving in one direction that each dz corresponds to.

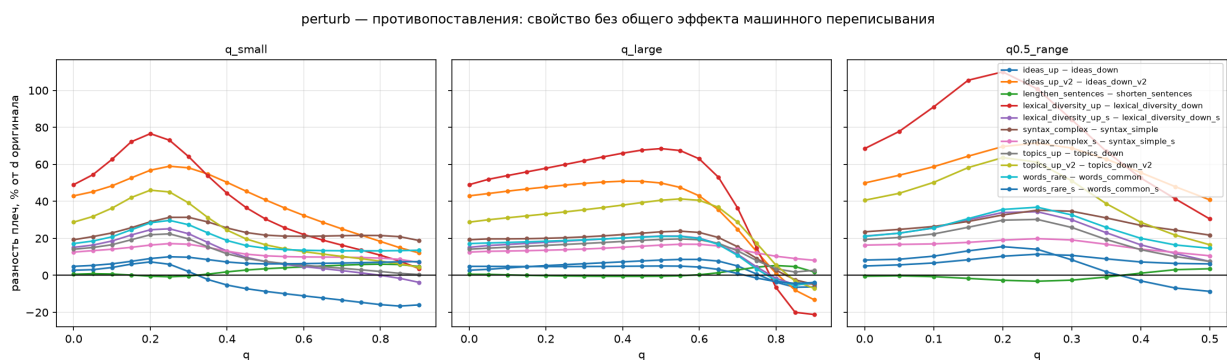


Figure 5. Each up-arm minus its down-arm, which cancels the component shared by all rewrites. In q_large both leading factors cross zero near $q = 0.8$ and turn negative: at the finest scale a more varied text measures as less dimensional. That is also the region where the power law itself breaks down, so it is recorded as an observation to be checked rather than a result.

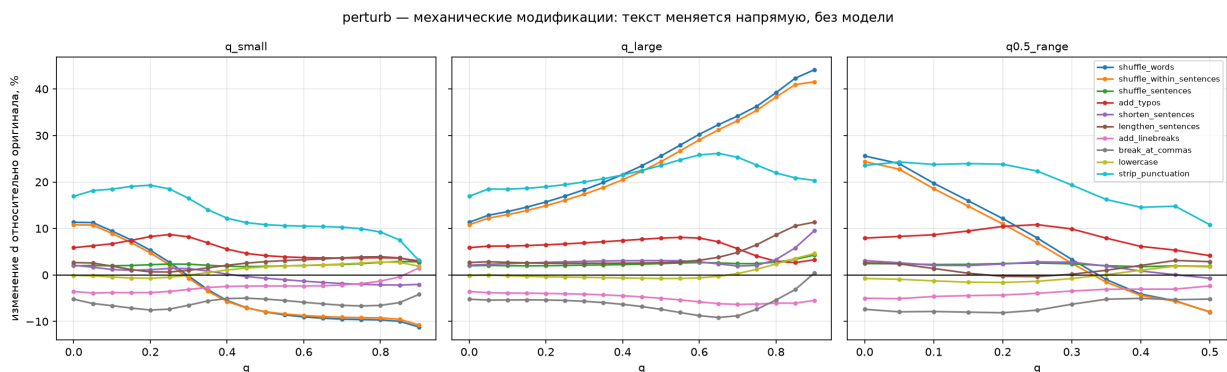


Figure 6. The mechanical perturbations alone. Three tiers are visible: shuffling and punctuation stripping at 10–44%, typos and line breaks at 2–9%, and sentence-length and sentence-order changes below 3% everywhere. Shuffling words across the text and shuffling them within sentences coincide almost exactly, while reordering whole sentences with word order intact is flat, which places the effect on word order inside a sentence rather than on the coherence of the text.

Four factors emerge with strong support. **Lexical diversity** is the largest, $dz = 2.43$, peaking in $q0.5_range$ at $q = 0.25$, reversing on both models and asymmetric: raising diversity moves d three times further than lowering it. Substituting rare words gives +25.7% where substituting common ones gives –7.2%, so the operative property is non-repetition rather than rarity. **Word order within a sentence** follows at $dz = 2.17$ under mechanical control, peaking in q_large at $q = 0.7$ and changing sign across the scale: –11.2% at $q_small = 0.9$ against +44.1% at $q_large = 0.9$. **Topic variety** reaches $dz = 1.93$ and **idea count** $dz = 1.82$, the latter being the only factor producing a large decrease, –47.5%, with an asymmetry opposite to the lexical one: human text sits near a ceiling in idea density but has room to fall.

Sentence length is not among them. It was the natural hypothesis, since texts at the low end of the dimension distribution proved to be dialogue and verse with short sentences, correlating with sentence count at –0.43. Manipulated directly, however, it moves d by about 1%, so the correlation was not causal; the $\pm 14\%$ seen through an LLM instructed to simplify syntax runs through the vocabulary it changes along the way. Typography behaves likewise, line breaks giving –3.8% and capitalisation –0.7%, though the latter had correlated at –0.51.

Peak locations divide the q axis cleanly. All five semantic properties peak in $q0.5_range$ between $q = 0.20$ and 0.35 ; all three structural ones peak in q_large between 0.7 and 0.8 , with typos in between. The axis appears to separate what is written from how it is arranged.

4.3 Do the two structural manipulations share a channel?

Punctuation stripping and word shuffling peak in the same regime at nearly the same q , and their effect profiles correlate at 0.61 , suggesting a single channel: the dismantling of local organisation. The suggestion is testable, since one shared resource implies that applying both lands near the larger of the two rather than near their sum. Two composite perturbations were run.

The hypothesis is rejected in the working range. At $q = 0.7$ the pair gives +67.2% where the larger alone gives +34.2%. Under a multiplicative null — effects on a ratio quantity composing as products — the two are independent to within 2 percentage points across $q = 0.5$ – 0.7 . That last statement depends on the null: a slope-additive null is exact at $q = 0$ and overshoots by 33 points at $q = 0.7$, and nothing in the geometry dictates which to use. Neither the rejection nor the saturation that appears by $q = 0.9$, where both nulls overpredict by 27 and 57 points, depends on that choice. The saturation is independent evidence that this regime measures local organisation specifically: once the first manipulation has run, the second has little left to destroy.

The mechanism behind punctuation stripping is known: the deleted tokens sit 1.5 times closer to their nearest neighbour than the rest, 11.5 against 17.4, and so carry the shortest MST edges that q_large sums over. This explains the effect rather than diminishing it. What the experiment does not establish is the wider claim that punctuation as a feature of style governs dimension, since the manipulation also fragments words; testing that would require substituting or inserting marks rather than deleting tokens.

5. Limitations

All results rest on one embedding model, ModernBERT-base, and the semantic fidelity metric is built on the same vectors as qPHD itself, so that check is not fully independent of what it verifies. The two rewriting models differ in force as well as in reliability — gemini moved type-token ratio by +40.5% where sonnet moved it +14.2%, both at full execution — so effect magnitudes are comparable only within a model. The sign reversal at $q_{\text{large}} \approx 0.8$ falls in the region where the power law itself degrades, with residuals of 4.3% against 0.3% in the working range, and requires separate verification. Finally, $d \propto L^{-0.21}$ does not converge within the reachable range of L , so absolute dimensions are properties of a measurement window and not of a text.

6. Reproduction

Code, per-text results, generated texts and figures are committed. The calibrated configuration is fixed in `scripts/config.py` with the measurement justifying each choice; the reference implementation is retained unmodified in `scripts/reference/` for line-by-line comparison, and the published settings remain reachable through a flag. The parameter sweeps that selected the configuration are separated into `experiments/`, indexed by the question each answers.